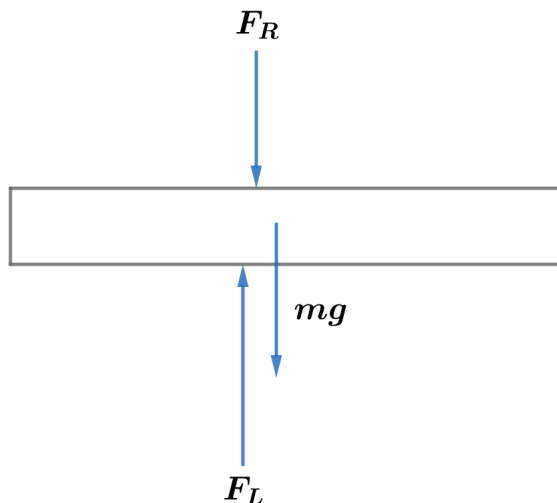


2019B F=ma Exam: Problem 8

Kevin S. Huang



We have

$$F_R + mg = F_L$$

so

$$F_R = F_L - mg = 200 \text{ N} - (5 \text{ kg})(10 \text{ m/s}^2) = 150 \text{ N}$$

This is the normal force at the top which the scale reads so it displays

$$m = \frac{F_R}{g} = \frac{150 \text{ N}}{10 \text{ m/s}^2} = 15 \text{ kg}$$

and the answer is A.